

Course: Practical Data Science I**Amna Rauf, Ye He, Reva Panda****Report for Policymakers regarding US opioids control****Executive Summary**

In 2010, the United States experienced a nation-wide epidemic of drug overdoses and deaths. Florida particularly was affected due to the rise of “pill mills”, clinics that prescribed large quantities of opioids. To combat these trends, Florida introduced strict policies to regulate the prescription and distribution of opioids in 2010. The aim of this project is to evaluate the effectiveness of this policy in reducing state level opioids drug prescriptions and overdose mortality rates with a difference in differences analysis. If Florida’s policy was effective, similar policies can be implemented across the United States to lower the burden of the opioid epidemic. Overall, the analysis showed that Florida’s policy was effective in reducing opioid prescriptions per capita and overdose mortality rates, proving that the adoption of similar policies would be effective in states like Nevada or North Carolina that have high rates of overdose mortality and drug shipment rates similar to Florida.

Discretionary Choices:

Difference in Difference was used instead of a simple pre-post analysis that compares only Florida’s data after policy implementation to data pre policy implementation to account for changes in opioid shipment and mortality trends that were caused by other phenomena outside of the policy implementation, such as natural decreases. Difference in Difference analysis isolates the impact of Florida’s policy

changes in 2010 by comparing the change in opioid shipment and overdose mortality data in Florida to states that did not enact similar policies. Florida's policy would be effective if the difference of drug overdose or prescription rate between Florida and the control states after the policy enactment is greater than the difference of drug overdose or prescription rate pre-policy enactment. Four control states were identified by a literature review of opioid prevalence and overdose rates to compare with Florida: Colorado, Nevada, North Carolina, and Florida. These states were chosen based on the two main criteria: 1) the control states did not enact similar policies to Florida from 2006 to 2014 and 2) each state experienced levels of opioid prescription and overdose similar to Florida prior to 2010. Their data after 2010 should best represent Florida's condition if there were no such policy being implemented in 2010 to demonstrate the comparison between policy effect and results without the policy. The three raw datasets that were used for this project were opioid shipment by county, number of deaths due to opioids overdose by county and county population data. Opioid shipment data were obtained from the Drug Enforcement Administration's ARCOS database, overdose mortality data from Centers for Disease Control and Prevention surveillance systems, and population data from the United States Census Bureau. Overall, the analysis showed that Florida's policy was effective in reducing opioid prescriptions per capita and overdose mortality rates.

Decisions to Be Made:

The policy, which required pain clinics to register with the state and banned them from dispensing drugs, reduced the rate of opioid prescriptions per capita in Florida, a trend that differed from the control states who did not enact anti-pill mill policies, proving

its effectiveness in reducing opioids prescriptions with a stricter requirement. After the policy implementation, Florida's mean opioid shipment rate decreased by 222.33 units from 2011 to 2014 while the control states opioid shipment rate increased by 35.39 units.

Additionally, policies regarding anti-pill mill similar to Florida could be implemented in states where the severity of overdose mortality was about the same level in Florida as our analysis shows that 2010 Florida policy was successful at reducing overdose mortality compared to control states. As presented by the data, Florida's mortality rate declined from 18.2 to approximately 13.4 *per 100,000* after policy implementation.

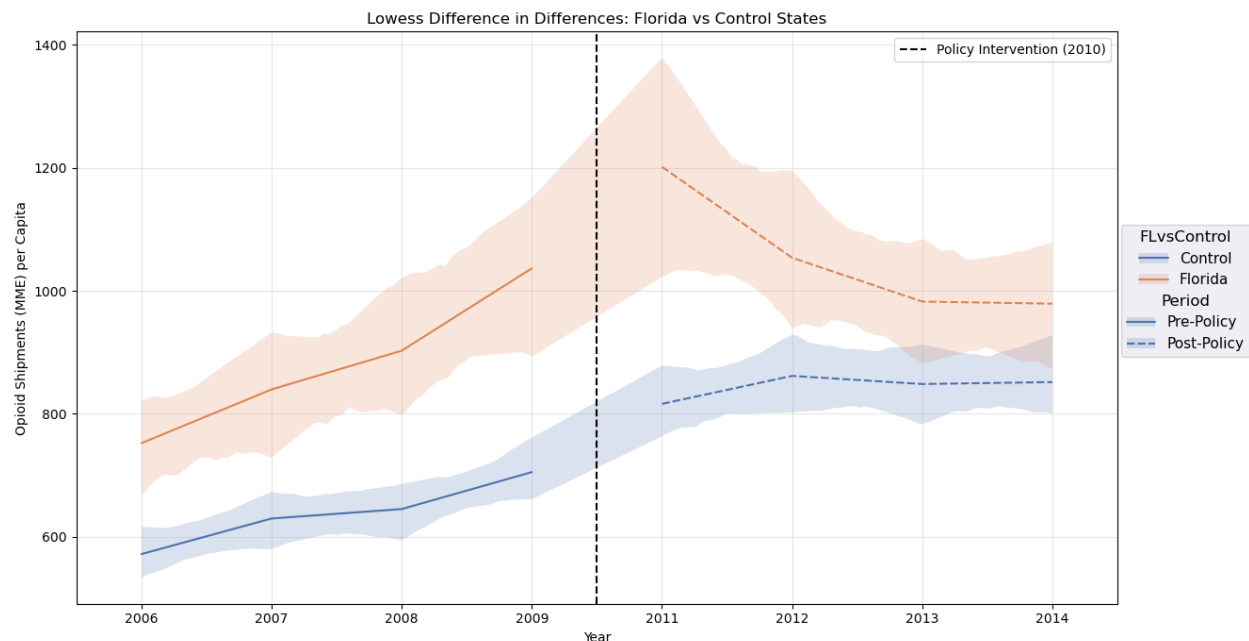
Despite the potential disadvantages of such policy like the possibility of turning to illegal markets for those already addicted to opioids, the quantitative comparison between Florida and control state proved the policy's effectiveness in reducing both opioids prescription and overdose mortality.

Results and Findings:

DID Analysis for Prescriptions:

All the prescription data were standardized into the same unit. In the analysis, we examined shipments of 14 different opioids including codeine, methadone, morphine, etc. Because these different drugs have different potencies, we converted the amount of each drug shipped to a common unit – Morphine Milligram Equivalents. Each county within each state had a drug prescription value that was calculated by a summation of all standardized potencies of the opioids shipped multiplied by its weight in milligrams, which was the final unit of our drug shipment data. All counties for each state are included, even

if the county received no drug shipments within the time of reference. Graphs showcase Drug Prescription per Capita.

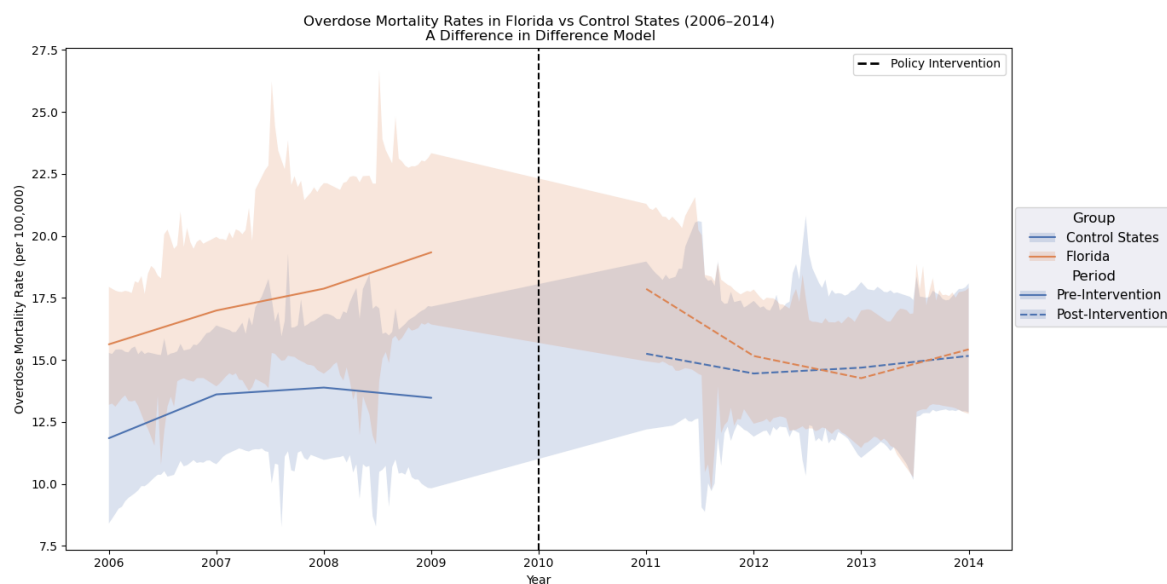


As seen in the chart above, Florida's drug policy was effective in reducing drug prescription rates. This graph shows the Prescriptions per Capita for Florida compared to an averaged line of the drug prescriptions rates of the 4 control states: Colorado, Nevada, North Carolina, and Wisconsin. Looking at the graph prior to 2010, when Florida enacted policies to reduce drug overdose and prescriptions, both Florida and the control states experienced a steady increase in drug prescription rates. While Florida did experience the highest levels of drug prescriptions per capita, the control states do provide valid comparison as they follow similar increases in drug prescription rates prior to 2010. Despite the policy being enacted in the beginning of 2010, the policy took a year to take effect, as seen by the peak in prescription per capita in 2011 in Florida. However, after 2011, there is a clear decrease in drug prescription rates. This differs from the other control

states. The control states experienced a leveling of drug shipment rates with no decrease. When performing a difference in difference analysis, the proportionate change of drug shipments between Florida and the control states is -31.99%. This shows that there was a decrease in the overall difference between mean drug prescription rates, indicating a causal relationship between Florida's drug policy and the decrease in drug prescriptions.

DID Analysis for Overdose Mortality:

In the data analysis of overdose mortality, the number of deaths due to overdose were summed for all the counties of a state while counties with a population less than 350,000 were excluded from the calculation. The overdose mortality was then calculated by the number of total deaths of all the included counties divided by the total population of these counties of a given year and then times 100,000. Then the number of deaths due to overdose by state per 100,000 population of a given year was obtained.



Similarly, the chart above shows that Florida's anti-pill-policy was effective in decreasing the overdose mortality rate in the state as well. Prior to 2010, overdose deaths of the

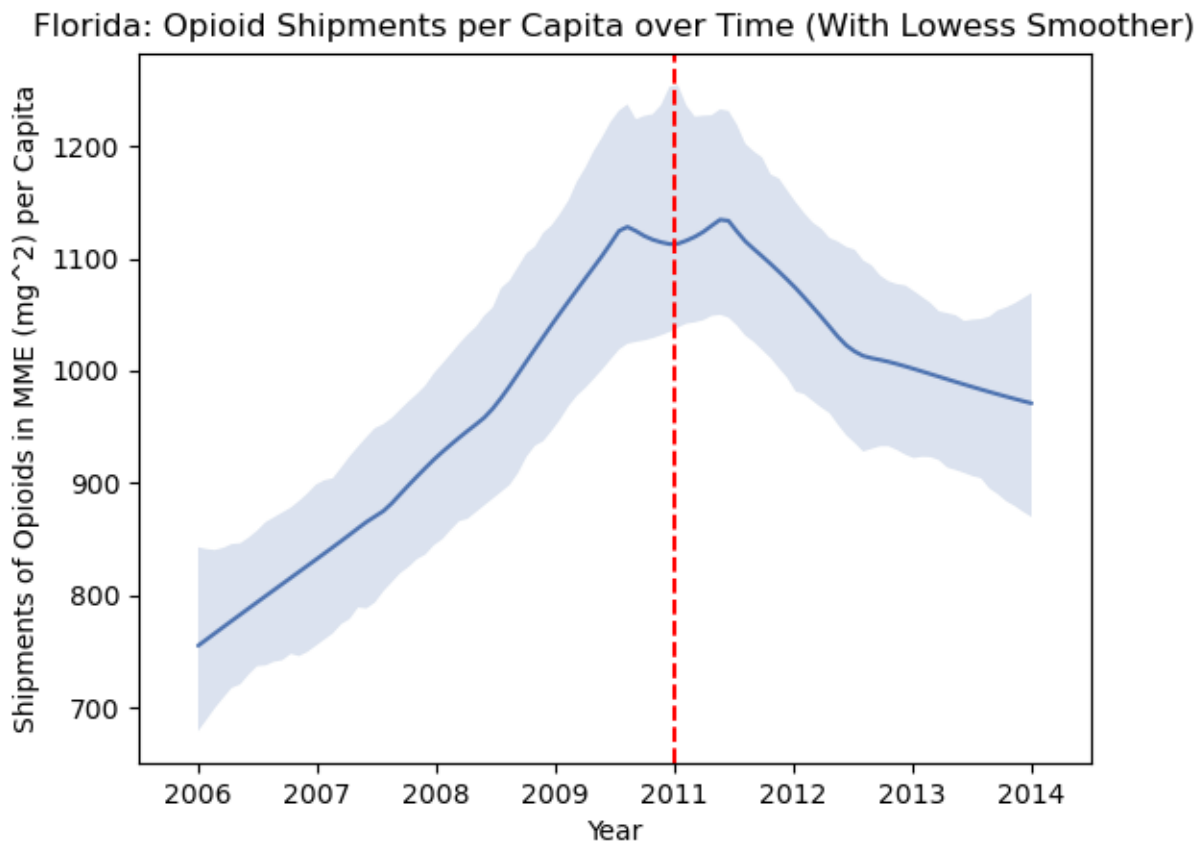
control states were rising at almost a steady and similar rate as Florida. Immediately after Florida implemented its policy in 2010, the trends split. Looking at the trends, Florida experienced a decrease in opioid mortality rates from 2011-2013, while the control states experienced a slight increase in deaths over this same period. This provides compelling evidence that the Florida policy reduced overdose mortality relative to what would have occurred without the intervention.

Policy Recommendations:

As shown above, Florida's policies did reduce drug prescriptions and overdose mortality rates, proving the effectiveness of strict pill-mill laws. The opioid crisis is a national health issue that requires intervention. Other states could enforce similar policies as Florida to reduce drug prescriptions and overdose mortality rates. Our analysis specifically revealed that high-risk states like Nevada which showed the highest mortality rates, which reached around 18.6 per 100,000 by 2014 should be prioritized for implemented Florida style policies. Additionally, while North Carolina did not have overall high rates of overdose mortality, the state continued to experience an increase in opioid prescription, indicating a need for stricter policies. Therefore, it is proposed that the federal government needs to create systems that track both prescription patterns and overdose deaths for real time evaluation. While Florida's approach is an excellent example of how strict pill-mill laws can work, the opioid crisis is complex. Therefore, the policies should try to find a balance between stopping drug abuse while also ensuring people with real pain can receive medications they require.

Appendix:

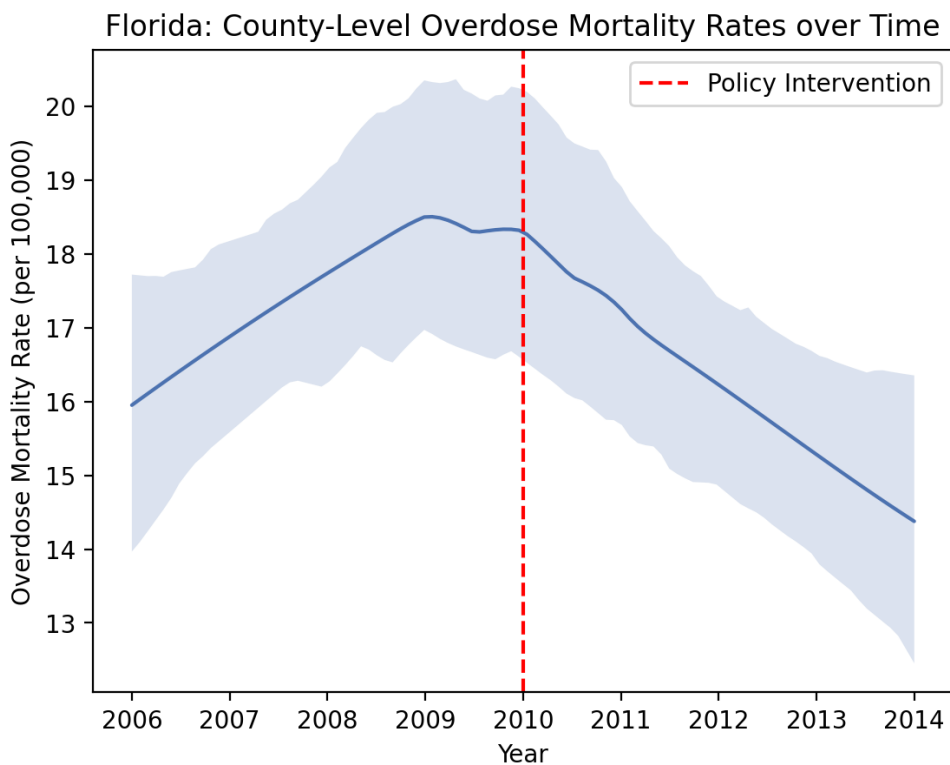
Pre-Post Graph for Florida Prescriptions



The Pre-Post Analysis analyzes the drug prescription per capita overtime. While overall the mean prescriptions per capita increased from 2006 to 2014 by 226.75, there was a decrease in the drugs prescribed per capita from 2011-2014 by **222.33**. The steady incline of drug prescriptions per capita ended shortly after Florida enacted its anti-pill-mill policy,

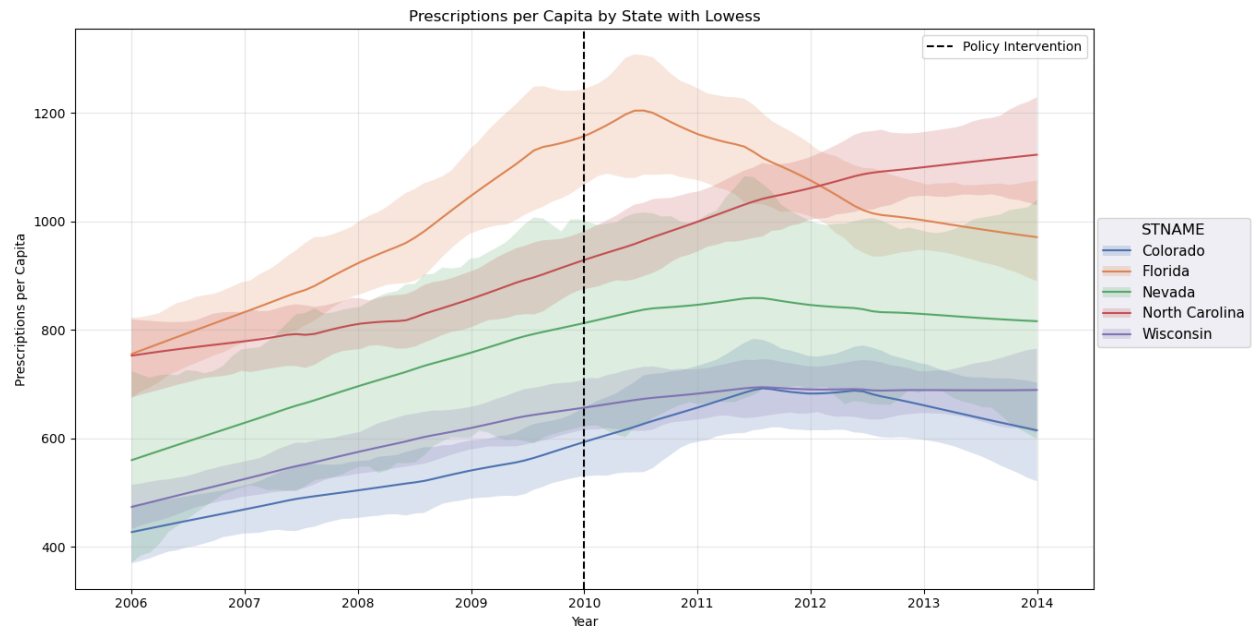
indicating that the policy might be effective. This is further proven in the Difference of Differences analysis.

Pre-Post Graph for Florida overdose mortality rates



The pre post analysis examines the overdose mortality per 100,000 persons over 2006 to 2014 in Florida. As shown above, overdose mortality rate in Florida was rapidly increasing from **14 per 100,000** in 2006 to **17 per 100,000** in 2009 but faces a steady decline from 2009 onwards showing that the implementation of the anti-mill-policy was indeed effective in Florida.

Difference in Difference Graph by State for Drug Shipment:



Difference in Differences by State for Overdose Mortality:

